Arduino Keypad and LCD Interface Project

Alireza Naderinasab

July 15, 2026

Contents

1 Project Overview

This project demonstrates interfacing a 4x4 matrix keypad with an Arduino Uno and a 16x2 LCD display.

The Arduino reads the pressed key from the keypad and displays the detected character on the LCD screen.

2 Required Components

- Arduino Uno
- 4x4 Matrix Keypad
- 16x2 LCD Display
- Breadboard
- Jumper Wires

3 Circuit Connections

3.1 LCD Connections

LCD Pin	Arduino Pin
RS	12
EN	11
D4	5
D5	4
D6	3
D7	2

3.2 Keypad Connections

Keypad Pin	Arduino Pin
Row 1	A0
Row 2	A1
Row 3	A2
Row 4	A3
Column 1	6
Column 2	7
Column 3	8
Column 4	9

4 Working Principle

The keypad works based on row and column scanning.

When a button is pressed, Arduino detects the corresponding row and column and displays the pressed key on the LCD.

5 Arduino Source Code

```

#include <Keypad.h>
#include <LiquidCrystal.h>

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);


const byte ROWS = 4;
const byte COLS = 4;


char keys[ROWS][COLS] = {
        {'1','2','3','A'},
        {'4','5','6','B'},
        {'7','8','9','C'},
        {'*','0','#','D'}
};


byte rowPins[ROWS] = {
        A0, A1, A2, A3
};


byte colPins[COLS] = {
        6, 7, 8, 9
};


Keypad keypad = Keypad(
makeKeymap(keys),
rowPins,
colPins,
ROWS,
COLS
);


void setup()
{
        lcd.begin(16,2);
        lcd.print("Press Key");
}


void loop()
{

        char key = keypad.getKey();
```

```
            if(key)
            {

                lcd.clear();

                lcd.setCursor(0,0);
                lcd.print("You Pressed:");

                lcd.setCursor(0,1);
                lcd.print(key);

            }

        }
```

6 Expected Output

After starting the system, LCD displays:

```
Press Key
```

The tested keys are:

- Key 4
- Key 6
- Key A
- Key B

Example outputs:

```
You Pressed:
4
```

```
You Pressed:
6
```

```
You Pressed:
A
```

```
You Pressed:
B
```

7 Project Images

Image 1

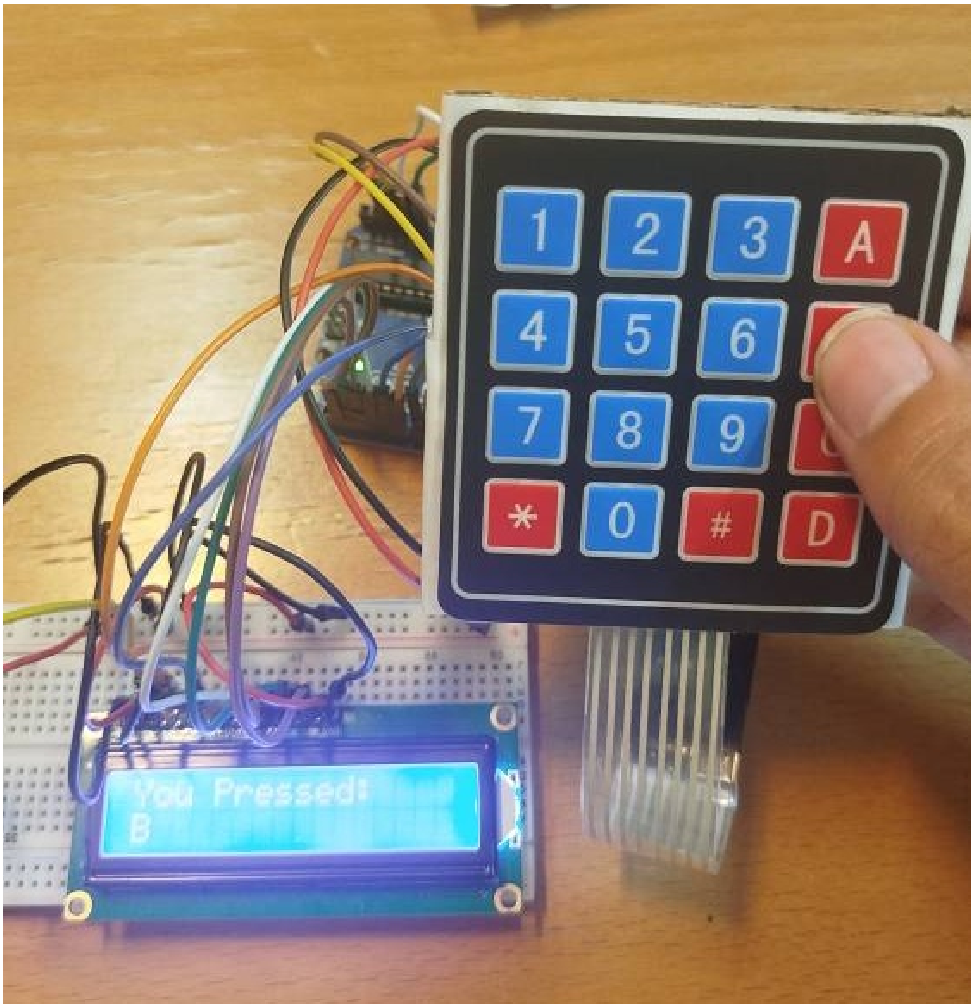


Image 2

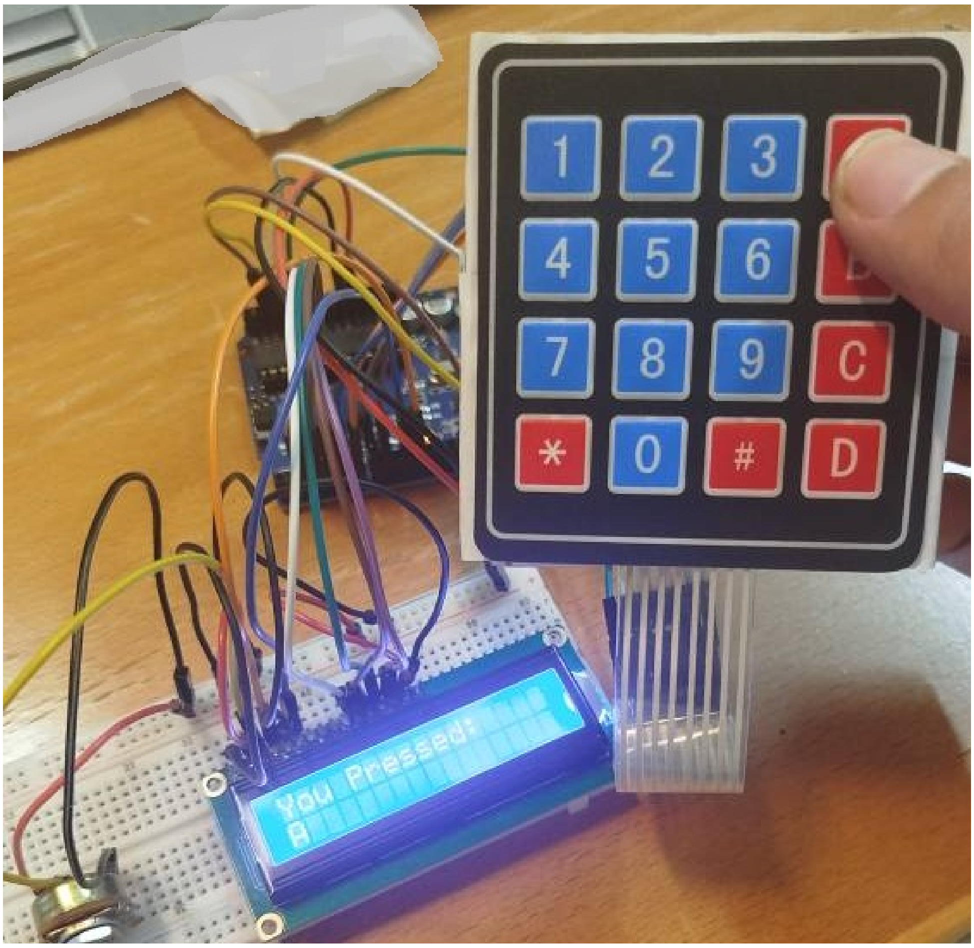


Image 3

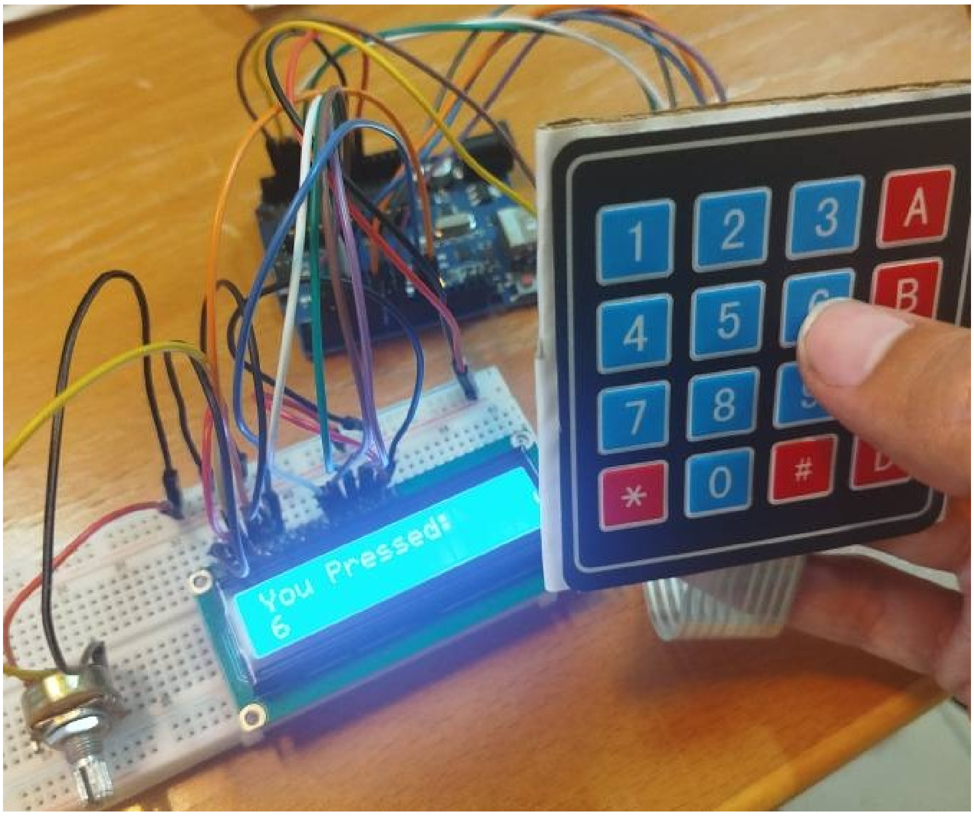
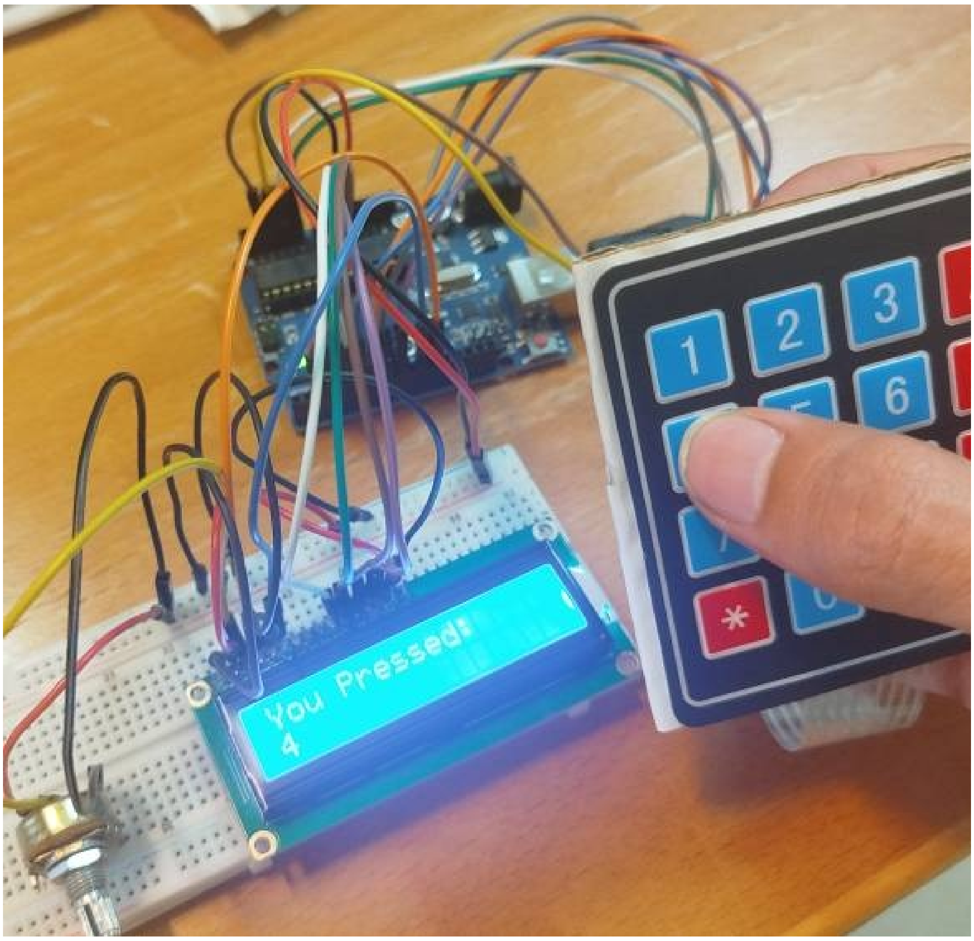


Image 4



8 Applications

- Password Security Systems
- Electronic Door Locks
- ATM Interfaces
- Calculator Systems
- Embedded User Interfaces

9 Conclusion

This project demonstrates communication between Arduino Uno, a 4x4 matrix keypad, and a 16x2 LCD display.

The system introduces keypad scanning, digital input detection, and LCD communication in embedded systems.